

Task 3: Customer Churn Prediction

Objective: Build a machine learning model to predict whether a customer will churn based on historical customer data.

Dataset: Churn_Modelling.csv

Steps Performed:

- Loaded the customer churn dataset.
- Removed irrelevant columns such as CustomerId, RowNumber, and Surname.
- Encoded categorical variables (Geography and Gender).
- Split the data into training and testing sets.
- Applied feature scaling using StandardScaler.
- Trained a Logistic Regression model.
- Evaluated the model using accuracy score, confusion matrix, and classification report.

Model Used: Logistic Regression

Conclusion: The model successfully predicts customer churn and can help businesses identify customers who are likely to leave, enabling proactive retention strategies.